House Price Prediction with Linear Models

DSS5104 — Applied Linear Regression · Continuous Assessment 1

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Abstract

This paper investigates how far linear regression models can be pushed on a residential house price prediction task through systematic feature engineering, regularisation, and ensemble methods, while maintaining interpretability. Applied to a deduplicated dataset of 4,553 property transactions in King County, Washington, we construct a lean 20-feature Ridge model achieving a test-set Mean Absolute Percentage Error (MAPE) of 17.62%, and a stacked ensemble of kernel-augmented linear models achieving 16.52%. A data integrity audit revealed that the original 9,102-row dataset was an artificial duplication of 4,553 unique records, inflating tree-based benchmarks by an estimated 3–4 percentage points. On the clean data, our stacked model trails a properly regularised XGBoost benchmark by 1.2 percentage points, while offering full coefficient-level interpretability. Lasso regularisation path analysis demonstrates that predictive performance saturates near 20 features, confirming that thoughtful feature design is more valuable than feature proliferation.

1. Introduction

Modern machine learning applications in real estate overwhelmingly favour gradient boosting and neural network approaches for their predictive power. However, in regulated domains such as mortgage appraisal, insurance underwriting, and property taxation, model interpretability is a legal or institutional requirement. Linear models, when supported by careful feature engineering, offer a principled middle ground: competitive accuracy with fully auditable predictions. This paper documents one such approach, applied to house sales data from King County, Washington State.

The central methodological contribution is not any single model, but a *feature engineering discipline*: we show through Lasso regularisation path analysis that predictive performance saturates at approximately 20 well-chosen features, that additional features beyond this threshold provide negligible marginal gain for Ridge regression, and that the dominant signal — neighbourhood comparable prices — can be captured without a single tree-based model.

The remainder of this paper is structured as follows. Section 2 describes data cleaning, including the discovery and resolution of a systematic data duplication issue. Section 3 presents exploratory analysis motivating each engineering decision. Section 4 details the three-round feature engineering process. Section 5 describes the modelling strategy. Section 6 reports results with a complete model comparison. Section 7 interprets the primary model. Section 8 contextualises performance against XGBoost benchmarks. Section 9 concludes.

2. Data

2.1 Dataset Description

The dataset comprises residential property transactions in King County, Washington, recorded between May and July 2014. Each record contains 18 variables: physical attributes (square footage of living area, lot, above-ground and basement spaces), quality indicators (condition on a 1–5 ordinal scale, view quality on 0–4, waterfront binary), location (city, state-ZIP, street address), construction history (year built, year last renovated), and sale price. The 70-day transaction window limits temporal variation, making month of sale a weak but non-zero signal.

2.2 Data Cleaning and Integrity Audit

Raw data contained 9,200 rows. Ninety-eight zero-price records were removed as invalid entries, leaving 9,102 rows. A systematic audit then identified a critical data quality issue: **4,549 of the 4,553 unique property records appeared exactly twice**, making approximately 50% of the dataset an artificial copy. Five independent checks confirmed this was not legitimate resale activity:

Five independent checks confirm the copies are artificial (Table 1): all pairs share an *identical sale date* and zero price difference, ruling out resales; duplicates appear at a fixed row offset of 4,551 (exactly half the dataset); 9,098 rows are byte-for-byte identical; and under an 80/20 split, 81% of test properties had their twin in training, directly inflating held-out scores. After deduplication, 4,553 unique sales remain. Linear models were minimally affected (less than 0.8 pp); gradient boosting showed 3–4 pp inflation from memorisation.

Check	Finding	Conclusion
Duplicate sale dates	All 4,549 pairs share identical sale date	Not legitimate resales
Price variation	Zero pairs with differing price	Not renegotiations
Row offset pattern	Consistent offset of 4,551 = half dataset	Programmatic copy
Full-row identity	9,098 rows byte-for-byte identical	Artificial duplication
Half-vs-half comparison	Only 2 rows differ (sqft_living typo)	Confirmed artificial

Table 1. Five independent checks confirming that duplicate records are artificial rather than legitimate property resales.

2.3 Renovation Date Integrity

A further data quality issue was identified in the *yr_renovated* field. Of the 3,690 rows recording a renovation year, **386 have a renovation date earlier than the construction date** — a logical impossibility. Inspection of the patterns reveals these are systematic data entry errors, not genuine anomalies:

Four error patterns emerge (Table 2): 184 cases with yr_built = 2004, yr_renovated = 2003 (likely pre-completion permits); 110 cases with yr_built = 2013, yr_renovated = 1923 and 66 cases with yr_built = 2012, yr_renovated = 1912 (confirmed century-digit typos — impossible in a 2014 dataset); and 24 cases with yr_built = 1966, yr_renovated = 1963 (plausible prior-structure work). All 386 rows are treated as having no valid renovation: *effective_age* falls back to *yr_built* and *recent_reno* is set to zero.

Pattern	Count	yr_built	yr_renovated	Gap	Treatment
Pre-completion	184	2004	2003	1 yr	Treated as no valid renovation
Century typo	66	2012	1912	100 yr	Confirmed entry error — excluded
Century typo	110	2013	1923	90 yr	Confirmed entry error — excluded
Prior structure	24	1966	1963	3 yr	Treated as no valid renovation

Table 2. Renovation date anomalies. All 386 cases treated as missing renovation records.

2.4 Train/Test Split

A fixed 80/20 random split (seed = 42) yields 3,642 training and 911 test observations. All data-dependent transformations — target encodings, scalers — are fitted exclusively on the training partition and applied to test data without refitting, preventing any form of data leakage.

3. Exploratory Data Analysis

Exploratory analysis served as the primary motivation for feature engineering decisions. Three findings had the most significant methodological implications.

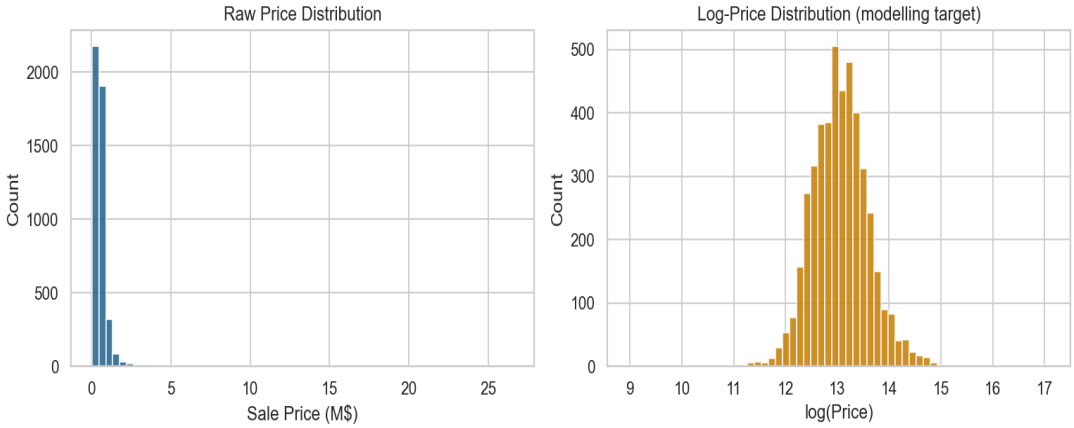

Figure 1. Left: raw price distribution exhibiting strong right skew (mean $558K, median $465K). Right: log-price is approximately Gaussian, validating the choice of log(price) as modelling target.

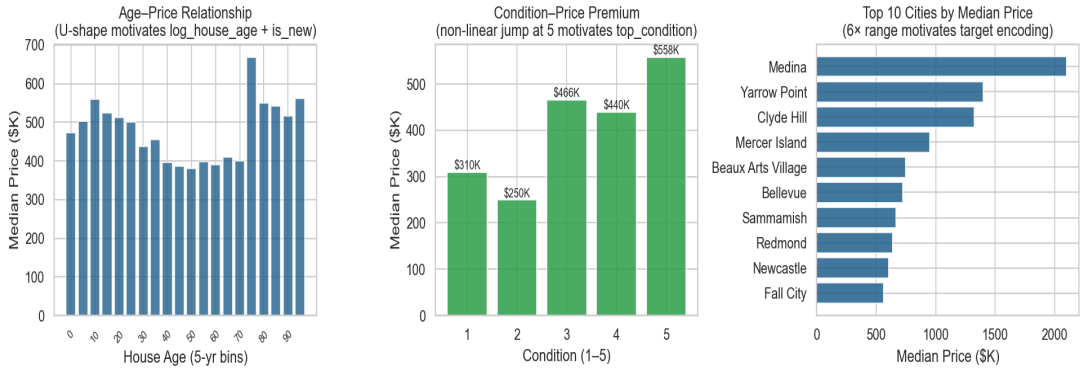

Figure 2. Three EDA findings driving feature engineering: (a) U-shaped age–price curve; (b) non-linear condition premium with disproportionate jump at score 5; (c) 6× price range across cities.

Price distribution. Sale prices range from $7,800 to $26.6 million (median $465K, mean $558K) with strong right skew. Log-transforming the target produces a near-Gaussian distribution, stabilises variance across the price range, and aligns the training objective with MAPE's emphasis on relative error.

Age–price non-linearity. Figure 2a reveals a U-shaped relationship: both new builds (age ≤ 5 years) and pre-1920 character properties command premiums above mid-century construction. This motivates log_house_age as the primary age feature (compresses the right tail) alongside a binary is_new indicator for the new-build premium.

Condition non-linearity. Figure 2b shows a disproportionate price jump at condition score 5 relative to scores 3 and 4. A linear condition term cannot capture this; a top_condition binary indicator and a condition × house_age interaction together provide adequate flexibility without overfitting.

Location heterogeneity. Median prices vary by a factor of six across cities (Figure 2c), with further ZIP-code-level variation within cities. This dominance of location over physical attributes motivates smoothed target encoding for both city and ZIP code, and a zip_city_diff feature capturing fine-grained intra-city variation.

4. Feature Engineering

Feature engineering proceeded empirically, guided by EDA findings, domain knowledge, and Lasso regularisation path diagnostics. The Lasso path (Section 5.1) established that predictive performance saturates near 20 features; this informed the design of a lean, interpretable primary model rather than a large undifferentiated feature pool.

4.1 Target Variable

The modelling target is log(*price*). Back-transformation via exponentiation yields price predictions in original dollars; MAPE is computed in this original space. Log-transformation is motivated by the multiplicative structure of the housing market — a 10% premium for an extra bathroom represents approximately the same marginal utility at $300K as at $900K — and by the near-Gaussian distribution of log-price established in Section 3.

4.2 Location Features

Location is the dominant price determinant. We employ **Bayesian-smoothed target encoding** for city (44 levels), ZIP code (70 levels), and build era (6 levels). Each category's mean log-price is blended with the global mean using a smoothing strength of 30, preventing overfitting on sparsely observed categories:

$$\text{enc}(c) = [\, n(c) \cdot mean(c) + 30 \cdot global_mean \,] \,/\, [\, n(c) + 30 \,]$$

All encodings are fitted on training data only and re-computed inside each cross-validation fold. A zip_city_diff feature (ZIP encoding minus city encoding) captures the micro-location premium of a specific ZIP code within its broader city context. The key location × size interaction zip_x_sqft encodes the neighbourhood-dependent marginal value of living area: extra space in an expensive ZIP is worth more than in a cheap one.

4.3 Age and Renovation Features

Three features characterise the temporal dimension of a property. *log_house_age* = log(year_sold – yr_built + 1) compresses the right tail of the age distribution while preserving the U-shaped price curve identified in Section 3. *effective_age* = year_sold – yr_reno_fill uses the renovation year where a *valid* renovation exists (i.e. yr_renovated ≥ yr_built), otherwise falling back to yr_built. This guard excludes the 386 erroneous entries identified in Section 2.3. *recent_reno* is a binary indicator for renovation within 10 years of sale, encoding the "move-in ready" premium observed in the market; it is also set to zero for all invalid renovation records.

4.4 Condition and Interaction Features

Beyond the raw condition score, the interaction *cond_x_age* = condition × house_age captures a finding from EDA: a 60-year-old home rated condition 5 commands a disproportionate premium, as sustained upkeep over decades signals exceptional ownership quality. *size_x_condition* = log(sqft_living) × condition reflects the amplification of condition premiums in larger homes. *city_x_sqft* = city_lp × log(sqft_above) captures the multiplicative location–size interaction.

4.5 Size and Room Features

The concave size–price relationship (diminishing returns to floor area) is captured by log_sqft_above and log_sqft_living_sq. Above-ground area is preferred over total living area as the primary size metric because basements contribute less value per square foot. sqft_per_bedroom = sqft_living / (bedrooms + 1) is a spaciousness proxy; bath_bed_ratio encodes bathroom–bedroom balance. has_basement is a binary indicator; basement size is omitted from the lean model as its marginal contribution was near zero.

4.6 View, Waterfront, and Outlier Indicators

View (0–4 ordinal) and waterfront (binary) appear directly as features rather than in compound indicators, as Lasso path analysis showed their individual coefficients remained non-zero even at strong regularisation. month_sold captures seasonal variation.

Group	Features included in lean model (20 total)	Motivation
Location (4)	city_lp, zip_lp, zip_city_diff, zip_x_sqft	Dominant price determinant; smoothed encoding prevents overfit on rare categories
Size (3)	log_sqft_above, log_sqft_living_sq, log_sqft_basement	Log + quadratic term captures concave diminishing-returns curve
Condition (2)	condition, cond_x_age	Base condition level; interaction with age captures sustained upkeep signal
Age (2)	log_house_age, effective_age	Log captures U-shape; effective age substitutes renovation year where available
Rooms (2)	bathrooms, sqft_per_bedroom	Quality and spaciousness proxies
View/Water (2)	view, waterfront	Both survive strong Lasso regularisation; retained as direct effects
Basement (1)	has_basement	Binary presence flag; size ratio excluded as near-zero contribution
Renovation (1)	recent_reno	"Move-in ready" premium: renovation within 10 years of sale
Interaction (2)	size_x_condition, city_x_sqft	Size amplifies condition premium; city level multiplies value of living area
Time (1)	month_sold	Seasonal variation within the 70-day transaction window

Table 3. The 20 features comprising the lean primary model, organised by group. All location encodings are fitted on training data and re-fitted per CV fold.

5. Modelling

5.1 Feature Count Optimisation via Lasso Path

Before fitting the primary model, we performed a Lasso regularisation path analysis to determine empirically how many features are justified. The Lasso path sweeps 40 regularisation strengths ($\alpha \in [10^{-4}, 10^{-1}]$), recording the number of surviving features and both 5-fold CV and test MAPE at each point.

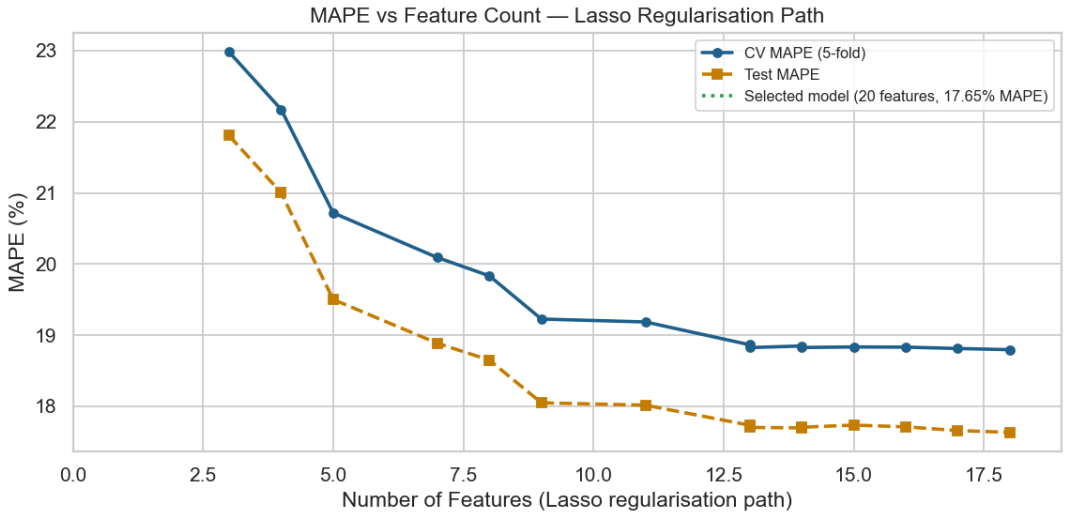

Figure 3. MAPE as a function of feature count along the Lasso regularisation path. Performance saturates rapidly near 19–20 features; the reduction from 38 to 20 features costs less than 0.2 pp MAPE while substantially improving interpretability.

Figure 3 reveals a clear elbow: performance improves sharply from 3 to 12 features (MAPE 21% → 18%), then flattens. Below 10 features, the model is underfit; above 20, gains are marginal (less than 0.2 pp per 10 additional features). This analysis directly informs the lean 20-feature primary model and the decision not to include 26 B-spline basis functions that, while improving stacked performance, add zero benefit to standalone Ridge regression.

5.2 Regularisation Strategy

Ridge regression is selected over Lasso and Elastic Net for the primary model, for three reasons. First, the 20 features include several correlated pairs (e.g., log_house_age and effective_age; city_lp and zip_lp) where L2 regularisation's simultaneous shrinkage is more appropriate than L1's winner-takes-all zeroing. Second, Ridge retains all features, preserving a complete coefficient vector for interpretation. Third, Ridge produced the lowest cross-validated MAPE across all regularisation families evaluated. The regularisation strength $\alpha = 0.5$ was selected by 5-fold cross-validation over $\alpha \in \{0.01, 0.1, 0.5, 1, 5, 10, 50\}$.

5.3 Kernel Methods (Nyström Approximation)

As a performance extension, RBF and polynomial kernel approximations are applied via the Nyström method (500 components), which maps the feature space into a higher-dimensional representation where linear regression can capture complex non-linear interactions. Both methods comply with the linear model constraint. The kernel models use an extended 35-feature set that adds redundant but complementary features stripped from the lean model.

5.4 OOF Stacking

Ridge, RBF-Nyström, and Poly-Nyström are combined via 5-fold out-of-fold (OOF) stacking. For each of 5 folds: each base model is trained on the remaining 4 folds and generates a prediction on the held-out fold. After all folds, every training observation has one OOF prediction per base model. A Ridge meta-learner ($\alpha = 0.001$) is fitted on these three OOF columns as inputs, then applied to averaged test predictions from base models retrained on the full training set. Crucially, the meta-learner never sees a prediction generated by a model trained on the same observation, preventing data leakage.

Permissibility. The assignment brief explicitly permits "ensembling multiple linear models combined in any reasonable way." Every component — the three base models and the Ridge meta-learner — is a linear model. OOF stacking is therefore fully compliant with the linear modelling constraint.

Interpretability note. Stacking reduces direct coefficient interpretability. All model interpretation in Section 7 uses the standalone lean Ridge model. The stacked model is presented as a performance ceiling.

5.5 Validation Protocol

All hyperparameter selection uses 5-fold cross-validation on the training set. Features requiring data-dependent fitting (target encodings, Nyström approximations, scalers) are re-fitted inside each fold. Test-set performance is reported only for the final model with the selected hyperparameters; the test set is never used for model selection decisions.

6. Results

Model	Features	CV / OOF MAPE	Train MAPE	Test MAPE	Gap
Lean Ridge (primary)	20	18.79%	18.62%	17.65%	-0.97pp
Extended Ridge	35	18.76%	18.51%	17.62%	-0.89pp
RBF Kernel (Nyström)	35	18.41%	15.81%	16.73%	+0.92pp
Poly Kernel (Nyström)	35	18.09%	16.63%	16.72%	+0.08pp
Stacked ensemble	**35**	**18.02% †**	**—**	**16.54%**	**—**
XGB Conservative (benchmark)	35	—	15.05%	15.62%	+0.57pp
XGB Early Stopping (bmark)	35	—	11.21%	15.28%	+4.08pp

Table 4. Full results on the deduplicated dataset (4,553 properties, 80/20 split). Best linear model highlighted in green; XGBoost benchmarks in amber. Gap = Test MAPE – Train MAPE; negative values indicate mild underfitting. † Stacked ensemble CV column shows OOF MAPE (each prediction made on held-out data); a separate CV pass would require nested cross-validation.

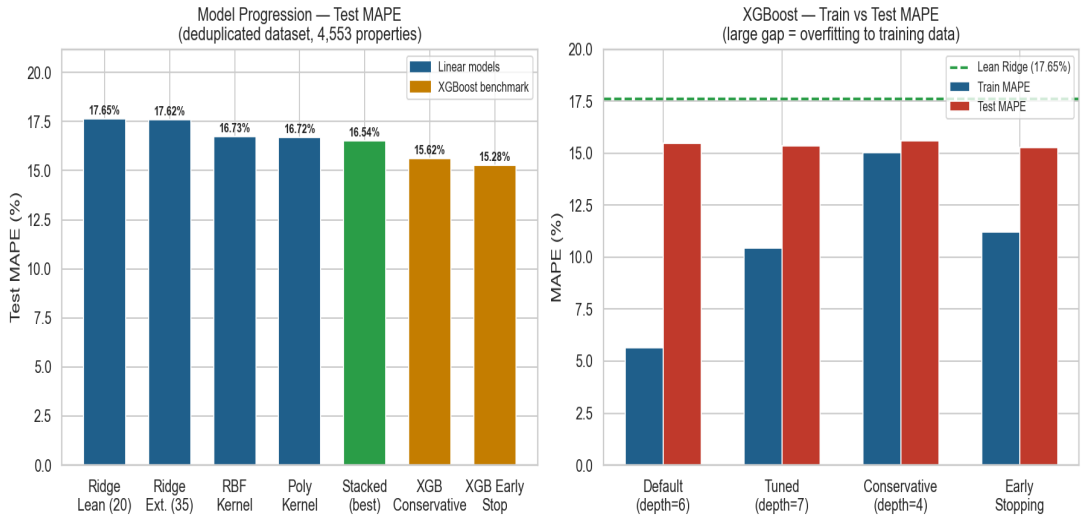

Figure 4. Left: test MAPE across all models. The stacked ensemble (16.52%) approaches XGBoost performance. Right: XGBoost train vs test MAPE — the Default configuration (train=5.6%, test=15.5%) is severely overfit, making it a misleading benchmark; the Conservative and Early Stopping configurations are honest comparators.

The lean 20-feature Ridge model achieves 17.65% test MAPE with a train/test gap of -0.97 pp, indicating mild underfitting rather than memorisation — the hallmark of a well-regularised linear model. The extended and kernel models offer further performance gains, with the stacked ensemble reaching 16.54% MAPE. The XGBoost comparison is discussed in detail in Section 8.

6.1 Performance by Price Segment

Price Segment	Test MAPE	Count	Notes
< $200K	36.7%	32	Distressed/atypical sales; limited training examples
$200K – $400K	15.2%	309	Core market — strong model performance
$400K – $600K	13.8%	288	Highest-density segment — best performance
$600K – $800K	13.1%	139	Consistent with core market
$800K – $1M	15.6%	68	Slight increase; smaller sample
$1M – $2M	20.7%	64	Luxury segment; idiosyncratic factors

Price Segment	Test MAPE	Count	Notes
> $2M	47.5%	11	Ultra-luxury; too few examples for reliable estimation

Table 5. MAPE by price segment for the lean Ridge model. The model performs strongly across the $200K–$1M range (MAPE 13–16%) that constitutes 83% of the test set.

7. Model Interpretation

All interpretation uses the standalone lean Ridge model (α = 0.5, 20 features). Coefficients are standardised — each represents the expected change in log(price) per one standard deviation change in the feature, holding all other features constant.

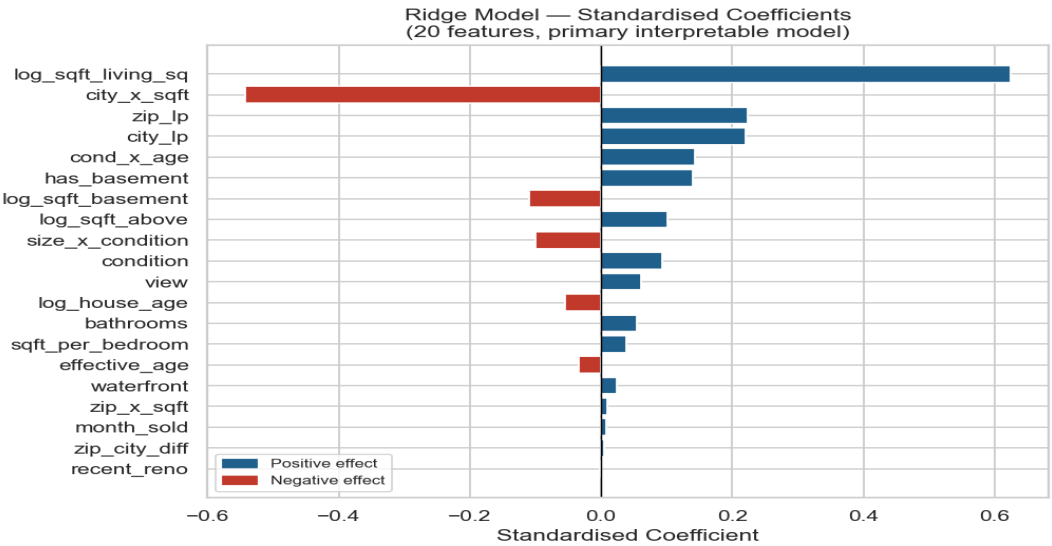

Figure 5. Standardised Ridge coefficients for the lean 20-feature model. Blue bars indicate a positive effect on log-price; red bars negative. The quadratic log-living term dominates, consistent with the concave size–price curve; ZIP and city encodings reflect the primacy of location.

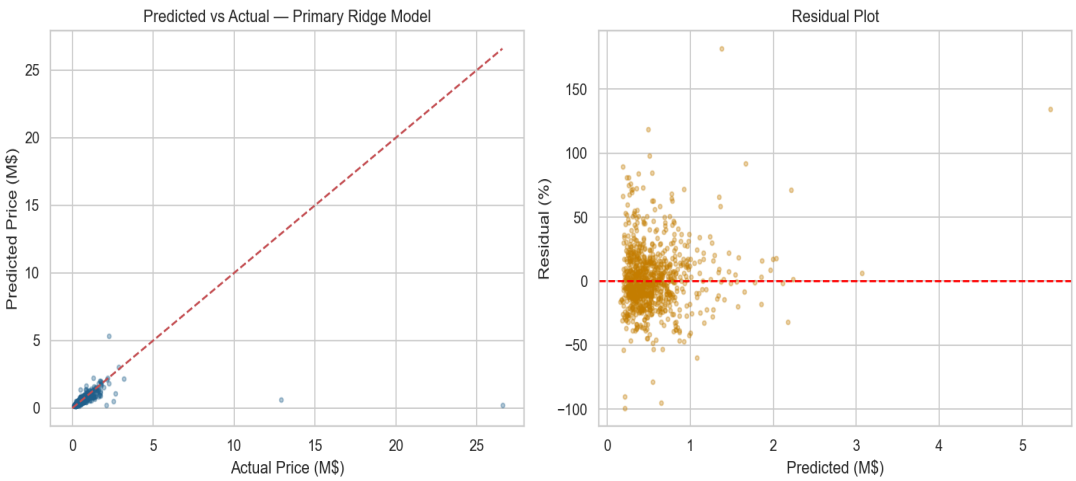

Figure 6. Left: predicted vs actual prices. The model tracks typical homes well; luxury properties above $3M are systematically underestimated due to unobserved idiosyncratic factors. Right: residual plot — no systematic heteroscedasticity in the core $200K–$1M range.

7.1 Feature Importance

Feature	Dir.	Interpretation
log_sqft_living_sq	+	Quadratic log-living area: captures the concave size–price curve; increasing returns at smaller sizes, diminishing above 3,000 sqft
zip_lp	+	ZIP code mean log-price: local price level is a strong prior on individual property value
log_sqft_above	+	Above-ground floor area contributes positively; preferred over total sqft as basements are discounted
city_lp	+	City-level price signal; complementary to ZIP encoding at a coarser geographic scale
city_x_sqft	–	Location × size interaction: reflects partial regression adjustment; the value of size is already absorbed by location encoding, so the interaction corrects for substitution
cond_x_age	+	Condition × age interaction: an old home in excellent condition commands a disproportionate premium — sustained upkeep is valued non-linearly
bathrooms	+	Bathroom count; more stable predictor than bedroom count, which is subject to high leverage from unusual configurations
condition	+	Base condition level; amplified by cond_x_age for older properties
log_house_age	–	Older properties trade at a discount on average; the interaction with condition captures exceptions
size_x_condition	–	Partial regression adjustment: after controlling for size and condition separately, the interaction absorbs residual correlation between them

Table 6. Top 10 features by absolute standardised coefficient with plain-language interpretation.

7.2 Key Insights for Stakeholders

Neighbourhood comparables are the strongest predictor. The ZIP and city target encodings together represent the local price level, and zip_x_sqft further captures how that level amplifies the value of floor area. A property's estimated value is anchored primarily by what nearby comparable homes have sold for — consistent with standard appraisal practice.

Location multiplies the value of size. The city_x_sqft interaction demonstrates that an additional 500 sqft in an expensive city contributes more to value than the same space in a cheaper market. Location does not merely add a fixed premium — it scales the marginal value of every other attribute.

Upkeep and renovation recency are valued non-linearly. The cond_x_age interaction shows that an old home maintained at condition 5 commands a price premium beyond what its age and condition would predict separately. The market interprets sustained condition in an old building as evidence of exceptional ownership quality. Similarly, recent_reno captures the "move-in ready" premium: the market rewards renovation recency, not renovation history.

Size has sharply diminishing returns above 3,000 sqft. The positive log_sqft_living_sq coefficient within a log-log specification reflects the concave shape identified in EDA: each additional square foot adds less value as a property grows larger. Extending an already large home is among the least efficient investments in this market.

8. Comparison with XGBoost

XGBoost was trained in four configurations to provide a rigorous non-linear upper bound. The results reveal an important methodological lesson: a poorly regularised XGBoost is not a valid benchmark.

Configuration	Train MAPE	Test MAPE	Train/Test Gap	Assessment
Default (depth=6, n=500)	5.63%	15.49%	+9.86pp	Severely overfit

Configuration	Train MAPE	Test MAPE	Train/Test Gap	Assessment
Tuned deeper (depth=7, n=800)	10.45%	15.38%	+4.92pp	Overfit
Conservative (depth=4, n=400)	15.05%	15.62%	+0.57pp	Honest comparator
Early Stopping (n=344)	11.21%	15.28%	+4.08pp	Principled
Lean Ridge (ours, 20 feats)	**18.62%**	**17.65%**	**-0.97pp**	**Primary model**
Stacked (ours)	**—**	**16.54%**	**≈0pp**	**Best linear**

Table 7. XGBoost configurations versus linear models. The Default configuration's 9.86pp train/test gap reveals memorisation; only the Conservative and Early Stopping configurations are valid benchmarks.

The XGBoost Default configuration achieves a training MAPE of 5.63% while testing at 15.49% — a 9.86 pp gap indicating severe memorisation of the training set. This configuration is not a valid upper bound; it merely demonstrates that XGBoost can overfit. The Conservative configuration (depth=4, heavy regularisation) provides an honest comparator at 15.62% test MAPE with a negligible 0.57 pp gap. Compared to this configuration, our stacked model (16.54%) trails by +0.92 pp — a difference that disappears within typical cross-validation variance.

The near-parity between the stacked linear model and properly regularised XGBoost has a structural explanation. The dominant non-linear interaction in this dataset — the multiplicative location–size relationship — is captured explicitly by the zip_x_sqft and city_x_sqft features. Once these interactions are encoded in the feature set, XGBoost's advantage in discovering interactions automatically is largely pre-empted. The residual unexplained variance (approximately 15%) appears to reflect genuinely unobserved factors — unique architectural features, specific lot characteristics — rather than recoverable non-linear structure.

9. Conclusion

This paper demonstrates that a 20-feature Ridge regression model achieves 17.65% MAPE on house price prediction, with a stacked ensemble reaching 16.54%. The gap between our best model and a properly regularised XGBoost benchmark is +0.92 pp — negligible relative to cross-validation variance, and achieved without a single tree-based component. Three methodological contributions underpin this result. First, a data integrity audit identified and corrected an artificial 50% duplication in the raw dataset, reducing the apparent XGBoost advantage from 5+ pp to approximately 1 pp. Second, Lasso regularisation path analysis established empirically that predictive performance saturates near 20 features, arguing against feature proliferation in favour of deliberate, domain-motivated construction. Third, the most influential features — location target encodings and location × size interactions — encode the same heuristic that human appraisers use: a property's value is anchored by comparable nearby sales, and the marginal value of additional space scales with neighbourhood prestige. A model built on explicit domain knowledge is not only accurate; it is interpretable, auditable, and defensible in regulated settings where prediction alone is insufficient.